ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

# TECHNICAL MATHEMATICS P1

## MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 10 Technical Mathematics Paper 1 practice examination.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Accept correct alternative methods; give follow-through marks (CA) where an earlier error is used correctly.

## QUESTION 1

**1.1** $\sqrt{9} = 3$: rational ✓; π: irrational ✓; $-\frac{3}{4}$: rational ✓; $\sqrt{-4}$: non-real ✓ (4)

**1.2** $45 = 32 + 8 + 4 + 1$ ✓ $= 101101_2$ ✓ (2)

**1.3** $16 + 8 + 0 + 2 + 0$ ✓ $= 26$ ✓ (2)

**1.4** $1011_2 + 110_2 = 10001_2$ ✓✓ $(11 + 6 = 17)$ (2)

**1.5** $3{,}142$ ✓ (1)

**1.6** $4{,}5$ ✓ $\times 10^{-5}$ ✓ (2)

**1.7** $\sqrt{49} = 7$ and $\sqrt{64} = 8$, so between 7 ✓ and 8 ✓ (2)

**[15]**

## QUESTION 2

**2.1.1** $8a^9b^6$ ✓✓ (2)

**2.1.2** $= \frac{x^3}{x^2}$ ✓✓ $= x$ ✓ (3)

**2.1.3** $4x^4$ ✓✓ (2)

**2.1.4** $2\sqrt{(4 \times 3)} + \sqrt{(9 \times 3)} = 4\sqrt{3}$ ✓ $+ 3\sqrt{3}$ ✓ $= 7\sqrt{3}$ ✓ (3)

**2.2.1** $(x - 3)(x - 4)$ ✓✓ (2)

**2.2.2** $(2x - 5)(2x + 5)$ ✓✓ (2)

**2.2.3** $3x(x + 2)$ ✓✓ (2)

**2.3** $= \frac{(x - 2)(x + 2)}{x + 2}$ ✓✓ $= x - 2$ ✓ (3)

**2.4** $2x^2 - 3x + 8x - 12$ ✓✓ $= 2x^2 + 5x - 12$ ✓ (3)

**2.5** $6x - 3 - 2x - 8$ ✓✓ $= 4x - 11$ ✓ (3)

**[25]**

## QUESTION 3

**3.1.1** $3x = 12$ ✓ $\rightarrow x = 4$ ✓ (2)

**3.1.2** $(x - 3)(x + 2) = 0$ ✓ $\rightarrow x = 3$ ✓ or $x = -2$ ✓ (3)

**3.1.3** $2^x = 2^5$ ✓ $\rightarrow x = 5$ ✓ (2)

**3.2** $-2x > 4$ ✓ $\rightarrow x < -2$ ✓ (sign reverses when dividing by a negative); number line with an open circle at $-2$ and arrow to the left ✓ (3)

**3.3** $3x + (2x + 1) = 11$ ✓ $\rightarrow 5x = 10$ ✓ $\rightarrow x = 2$ ✓; $y = 2(2) + 1 = 5$ ✓ (4)

**3.4** $R = \frac{V}{I}$ ✓ $= \frac{12}{0,5}$ ✓ $= 24\ \Omega$ ✓ (3)

**3.5** $30 = 2(9 + b)$ ✓ $\rightarrow 15 = 9 + b$ ✓ $\rightarrow b = 6$ cm ✓ (3)

**3.6** Let width $= x$: $x(x + 3) = 40$ ✓ $\rightarrow x^2 + 3x - 40 = 0$ ✓ $\rightarrow (x + 8)(x - 5) = 0$ ✓ (5)
$x = 5$ (width cannot be negative) ✓; width 5 cm and length 8 cm ✓

**[25]**

## QUESTION 4

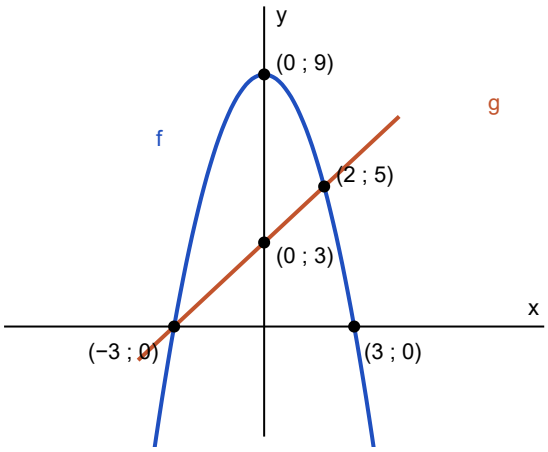
**4.1** $(0\ ;\ 9)$ ✓ (1)

**4.2** $-x^2 + 9 = 0 \rightarrow x = \pm 3$ ✓; $(-3\ ;\ 0)$ and $(3\ ;\ 0)$ ✓ (2)

**4.3** $x + 3 = 0 \rightarrow (-3\ ;\ 0)$ ✓ (1)

**4.4** $x = 0$ ✓; gradient $= 1$ ✓ (2)

**4.5** (5)




Marks: shape of *f* (opens downwards) ✓; turning point (0 ; 9) ✓; *x*-intercepts of *f* ✓; straight line *g* with positive gradient ✓; intercepts of *g*: (0 ; 3) and (−3 ; 0) ✓

**4.6** $-x^2 + 9 = x + 3 \rightarrow x^2 + x - 6 = 0$ ✓ $\rightarrow (x + 3)(x - 2) = 0$ ✓ (4)
$(-3\ ;\ 0)$ ✓ and $(2\ ;\ 5)$ ✓

**4.7** $y \leq 9$ ✓ (1)

**4.8** $-3 \leq x \leq 2$ ✓✓ (2)

**4.9** $x = 0$ ✓ and $y = 0$ ✓; quadrants 1 and 3 ✓ (3)

**4.10** $(0\ ;\ 1)$ ✓; $y = 0$ ✓ (2)

**4.11** $y = x - 2$ ✓✓ (2)

**[25]**

## QUESTION 5

**5.1** (a) $A = P(1 + i)^n = 15\,000(1{,}09)^4$ ✓✓ $= R21\,173{,}72$ ✓✓ (7)

(b) $A = P(1 + in) = 15\,000(1 + 0{,}09 \times 4)$ ✓✓ $= R20\,400$ ✓

**5.2** $21\,173{,}72 - 20\,400 = R773{,}72$ ✓ (1)

**5.3** $A = 40\,000(1 - 0{,}1 \times 3)$ ✓ $= R28\,000$ ✓ (2)

**[10]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**